

# How Long Reasoning Chains Influence LLMs' Judgment of Answer Factuality

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## Abstract

Large language models (LLMs) are widely used as scalable surrogates for human evaluation, yet such judges remain susceptible to surface-level biases. One possible reason is that these judges lack sufficient evidence in assessing answer correctness. With the rise of reasoning-capable models, exposing a generator's reasoning chain to the judge is a natural candidate for improving judgment accuracy, but its actual impact remains understudied. In this paper, we systematically investigate how access to reasoning chains affects LLM-based judgment on factual question answering and mathematical reasoning. We find that weak judges are easily swayed by reasoning presence, frequently accepting incorrect answers accompanied by fluent reasoning, while strong judges can partially leverage reasoning as informative evidence. Nevertheless, even strong judges are misled by seemingly high-quality reasoning chains. These findings highlight the need for more robust LLM judges that can distinguish genuine reasoning quality from superficial fluency when evaluating modern reasoning models.

## Keywords

LLM-as-a-Judge, Reasoning Chains, Chain-of-Thought

## 1 Introduction

Reliable evaluation is fundamental to understanding the capabilities of AI models and guiding their future development. For open-ended tasks, human evaluation is widely regarded as the gold standard, yet it is costly, time-consuming, and difficult to scale [3, 5]. This limitation has motivated a rapidly growing line of work that adopts strong LLMs as scalable surrogate evaluators, with judgments shown to correlate closely with human preferences [10, 14]. Despite their growing adoption, LLM-based judges remain imperfect: they are sensitive to surface-level features such as answer length, fluency, or phrasing, and may struggle to distinguish correct answers from plausible but incorrect ones [4, 11]. One possible reason is that, given only a single generated answer, the judge lacks sufficient information to reliably assess correctness.

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| Misleading Susceptibility                                                                                                                                                                                                                                                                               | Effective Utilization                                                                                                                                                                                                                                                                                        |        |        |           |  |  |  |  |  |          |  |  |                                                                                                                                                                                            |  |      |        |           |  |  |  |  |  |          |  |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|-----------|--|--|--|--|--|----------|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------|--------|-----------|--|--|--|--|--|----------|--|--|
| <b>Prompt:</b> Judge whether the following answer about the question is correct.                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                              |        |        |           |  |  |  |  |  |          |  |  |                                                                                                                                                                                            |  |      |        |           |  |  |  |  |  |          |  |  |
| <b>Question:</b> Who is the girl in green day 21 guns?<br><b>Model Response:</b> Amanda Kessel<br><b>Ground Truth:</b> Lisa Stelly                                                                                                                                                                      | <b>Question:</b> When did Toyota first come to the united states?<br><b>Model Response:</b> 1958<br><b>Ground Truth:</b> 1957                                                                                                                                                                                |        |        |           |  |  |  |  |  |          |  |  |                                                                                                                                                                                            |  |      |        |           |  |  |  |  |  |          |  |  |
| <b>Reasoning Chain:</b> I remember that "21 Guns" is a song by Green Day from their album "American Idiot." (...) Maybe a fan who was part of the video shoot?n/n I believe her name is Amanda Kessel. (...) The user probably wants the name of the girl in the video, so the answer is Amanda Kessel. | <b>Reasoning Chain:</b> I remember that Toyota started exporting cars to the US in the 1950s. Yes, the first Toyota car sold in the US was the Toyopet Corona in 1958. (...) But some sources might mention 1957 for the initial import. However, the commonly accepted answer is 1958. Let me go with that. |        |        |           |  |  |  |  |  |          |  |  |                                                                                                                                                                                            |  |      |        |           |  |  |  |  |  |          |  |  |
| <table><tr><td></td><td>Weak</td><td>Strong</td></tr><tr><td>w/o Think</td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td>w/ Think</td><td></td><td></td></tr></table>                                                                                                              |                                                                                                                                                                                                                                                                                                              | Weak   | Strong | w/o Think |  |  |  |  |  | w/ Think |  |  | <table><tr><td></td><td>Weak</td><td>Strong</td></tr><tr><td>w/o Think</td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td>w/ Think</td><td></td><td></td></tr></table> |  | Weak | Strong | w/o Think |  |  |  |  |  | w/ Think |  |  |
|                                                                                                                                                                                                                                                                                                         | Weak                                                                                                                                                                                                                                                                                                         | Strong |        |           |  |  |  |  |  |          |  |  |                                                                                                                                                                                            |  |      |        |           |  |  |  |  |  |          |  |  |
| w/o Think                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                              |        |        |           |  |  |  |  |  |          |  |  |                                                                                                                                                                                            |  |      |        |           |  |  |  |  |  |          |  |  |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                              |        |        |           |  |  |  |  |  |          |  |  |                                                                                                                                                                                            |  |      |        |           |  |  |  |  |  |          |  |  |
| w/ Think                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                              |        |        |           |  |  |  |  |  |          |  |  |                                                                                                                                                                                            |  |      |        |           |  |  |  |  |  |          |  |  |
|                                                                                                                                                                                                                                                                                                         | Weak                                                                                                                                                                                                                                                                                                         | Strong |        |           |  |  |  |  |  |          |  |  |                                                                                                                                                                                            |  |      |        |           |  |  |  |  |  |          |  |  |
| w/o Think                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                              |        |        |           |  |  |  |  |  |          |  |  |                                                                                                                                                                                            |  |      |        |           |  |  |  |  |  |          |  |  |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                              |        |        |           |  |  |  |  |  |          |  |  |                                                                                                                                                                                            |  |      |        |           |  |  |  |  |  |          |  |  |
| w/ Think                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                              |        |        |           |  |  |  |  |  |          |  |  |                                                                                                                                                                                            |  |      |        |           |  |  |  |  |  |          |  |  |

Figure 1: Examples from NQ showing that visible reasoning can mislead weak judges but help strong judges identify errors. “w/o Think” uses only the final answer, whereas “w/ Think” also includes the reasoning chain.

Recent advances have endowed LLMs with explicit reasoning capabilities, enabling them to produce step-by-step thinking processes prior to arriving at a final answer. Beyond the answer itself, these visible reasoning traces offer LLM judges a richer signal for evaluation—yet whether exposing such reasoning actually improves judgment quality remains an open question. Inspired by human decision-making, we further ask whether models differ in how they use such reasoning. Humans with limited expertise may be persuaded by fluent but incorrect explanations, whereas experts can leverage reasoning as evidence to scrutinize correctness. We therefore suspect that weak LLM judges may over-trust the presence of reasoning, while stronger ones may be better positioned to treat it as informative evidence rather than a persuasive signal.

**Table 1: LLM-as-a-Judge evaluation results under the Qwen3-8B answer generator.**

| Generator | Dataset | Acc  | Judge Models      | Alignment   |             | Pass Rate   |             | Overconfidence |             | Conservativeness |             |
|-----------|---------|------|-------------------|-------------|-------------|-------------|-------------|----------------|-------------|------------------|-------------|
|           |         |      |                   | w/o Think   | w/ Think    | w/o Think   | w/ Think    | w/o Think      | w/ Think    | w/o Think        | w/ Think    |
| Qwen3-8B  | NQ      | 23.2 | Qwen3-8B          | 58.2        | 33.2        | 57.8        | 88.0        | 38.2           | 65.8        | <b>3.6</b>       | 1.0         |
|           |         |      | Qwen3-14B         | 58.0        | 36.4        | 58.0        | 84.0        | 38.4           | 62.2        | <b>3.6</b>       | 1.4         |
|           |         |      | Llama3-8B         | 38.6        | 28.4        | <b>79.8</b> | <b>93.2</b> | <b>59.0</b>    | <b>70.8</b> | 2.4              | 0.8         |
|           |         |      | GPT-4o            | 74.0        | 74.0        | 44.0        | 41.2        | 23.4           | 22.0        | 2.6              | 4.0         |
|           |         |      | DeepSeek-v3.1     | 63.4        | 76.2        | 55.8        | 35.4        | 34.6           | 18.0        | 2.0              | 5.8         |
|           |         |      | Claude Sonnet 4.5 | <b>75.8</b> | <b>84.0</b> | <b>79.8</b> | <b>93.2</b> | <b>59.0</b>    | <b>70.8</b> | 2.2              | <b>10.2</b> |
|           | MATH500 | 84.8 | Qwen3-8B          | 85.0        | 86.6        | 86.6        | 97.8        | 8.4            | 13.2        | 6.6              | 0.2         |
|           |         |      | Qwen3-14B         | <b>88.6</b> | 86.8        | 84.2        | 98.0        | 5.4            | 13.2        | 6.0              | 0.0         |
|           |         |      | Llama3-8B         | <b>88.6</b> | 88.8        | 92.2        | 83.6        | 9.4            | 5.0         | 2.0              | 6.2         |
|           |         |      | DeepSeek-v3.1     | 86.0        | 85.8        | <b>98.4</b> | <b>98.2</b> | <b>13.8</b>    | <b>13.8</b> | 0.2              | 0.4         |
|           |         |      | GPT-4o            | 86.8        | 88.8        | 74.0        | 76.4        | 1.2            | 1.4         | <b>12.0</b>      | 9.8         |
|           |         |      | Claude Sonnet 4.5 | 77.8        | <b>89.4</b> | 94.6        | 76.6        | 10.2           | 7.0         | 0.4              | <b>15.2</b> |

**Table 2: LLM-as-a-Judge evaluation results under the DeepSeek-V3.1 answer generator.**

| Generator     | Dataset | Acc  | Judge Models      | Alignment   |             | Pass Rate   |             | Overconfidence |             | Conservativeness |             |
|---------------|---------|------|-------------------|-------------|-------------|-------------|-------------|----------------|-------------|------------------|-------------|
|               |         |      |                   | w/o Think   | w/ Think    | w/o Think   | w/ Think    | w/o Think      | w/ Think    | w/o Think        | w/ Think    |
| DeepSeek-V3.1 | NQ      | 46.4 | Qwen3-8B          | 75.8        | 50.6        | 46.6        | 94.2        | 12.2           | 48.6        | <b>12.0</b>      | 0.8         |
|               |         |      | Qwen3-14B         | 75.8        | 49.0        | 54.2        | 95.4        | 16.0           | 50.0        | 8.2              | 1.0         |
|               |         |      | Llama3-8B         | 77.8        | 47.0        | <b>61.0</b> | <b>98.6</b> | <b>18.4</b>    | <b>52.6</b> | 3.8              | 0.4         |
|               |         |      | GPT-4o            | 88.0        | 51.6        | 52.8        | 91.6        | 9.2            | 46.8        | 2.8              | 1.6         |
|               |         |      | DeepSeek-V3.1     | 80.4        | 49.0        | 46.0        | 95.8        | 9.6            | 50.2        | 10.0             | 0.8         |
|               |         |      | Claude Sonnet 4.5 | <b>90.0</b> | <b>55.8</b> | 48.8        | 69.8        | 6.2            | 33.8        | 3.8              | <b>10.4</b> |

To answer these questions, we conduct a systematic study of LLM-based judgment on both factual QA and mathematical reasoning tasks, comparing judge behavior with and without access to the generator’s reasoning chain across judges of different capability levels. Our results show that weak judges are easily swayed by the mere presence of fluent reasoning and frequently accept incorrect answers, whereas strong judges are more selective. They can, in some cases, identify errors within the reasoning and use this to better assess correctness. However, they can still be misled by seemingly high-quality chains. These findings highlight the need to move beyond answer-only evaluation paradigms and call for more robust judge designs.

## 2 Experimental Setup

**Models.** We use Qwen3-8B [13] and DeepSeek-V3.1 [6] for answer generation. For judgment, six models are employed: Qwen3-8B, Qwen3-14B, Llama-3.1-8B [1], DeepSeek-V3.1, GPT-4o [8], and Claude Sonnet 4.5 [2], where the latter three are considered strong models for their strong QA performance and the rest weak.

**Datasets.** To examine the impact of task type, we conduct experiments on two benchmarks: Natural Questions (NQ) [9], which focuses on factual question answering, and MATH-500 [7], which focuses on mathematical reasoning. For each benchmark, we randomly sample 500 questions to manage computational cost.

**Evaluation Metrics.** Following [12], we adopt **Accuracy** as the metric for QA performance evaluation, and employ four metrics for judge performance evaluation: (1) **Alignment**, the proportion of cases in which the judge’s verdict agrees with the ground-truth correctness label; (2) **Pass Rate**, the proportion of cases in which the judge deems a response correct; (3) **Overconfidence**, the proportion of incorrect responses that are mistakenly judged as correct; and (4) **Conservativeness**, the proportion of correct responses that are erroneously judged as incorrect.

## 3 Results and Analysis

Judge performance results are presented in Table 1 and Table 2, with conclusions summarized as follows.

**(1) Weak judges tend to be misled by reasoning chains, while strong judges remain selective and can even benefit from them.** As shown in Table 1, on both NQ and MATH500, the pass rate of weak judges increases substantially after observing the reasoning process, notably exceeding the actual answer accuracy and leading to severe overconfidence. This suggests that weak models are generally misled by fluent but potentially incorrect reasoning. In contrast, for strong judges, exposure to the reasoning process can sometimes reduce pass rates while improving alignment, indicating that they are capable of identifying errors within the reasoning.

**(2) Even strong models can be misled by seemingly high-quality reasoning.** As shown in Table 2, on NQ, the reasoning chains generated by DeepSeek-V3.1 cause even strong models to substantially increase their pass rate, resulting in severe overconfidence. We conjecture that this arises because, compared to Qwen3-8B, the reasoning chains generated by DeepSeek-V3.1 appear higher in quality and are consequently more likely to mislead.

## 4 Conclusion

In this paper, we ask whether exposing the generator’s reasoning chain improves the reliability of LLM-based judges. Through a systematic study on both factual QA and mathematical reasoning tasks, we find that reasoning is a double-edged sword: weak judges are easily swayed by the presence and fluency of reasoning and frequently accept incorrect answers, whereas strong judges are more selective and can sometimes identify errors within the chain, yet they remain vulnerable to seemingly high-quality but misleading reasoning. These findings highlight the limitations of answer-only evaluation paradigms and call for more robust judge designs that can critically verify, rather than be persuaded by, reasoning chains.

## Acknowledgments

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